

WARNING!

Dear students,

- 1- The final exam will be released as homework on Tuesday at 13:15.
- 2- Homework must be submitted on the same day.
- 2- The deadline for submission of the assignment is 17:00. Upload your homework before this time.
- 3- No additional time will be given.
- 4- Submit the homework as a single pdf. Please do not upload your homework as a few pictures.
- 5- Upload your answers as a single file.
- 6- Let the name of the file you uploaded include your name, surname and number.

Sayın öğrenciler,

- 1- Final sınavı, ödev olarak Salı günü saat 13:15 te serbest bırakılacaktır.
- 2- Ödev aynı gün teslim edilmesi gereklidir.
- 3- Ödevin son teslim saati 17:00 dır. Busesaaten önce ödeviniz yükleyiniz.
- 4- Ek süre verilmeyecektir.
- 5- Ödevleri, tek bir pdf olarak teslim ediniz. Lütfen, ödevinizi birkaç resim olarak yüklemeyiniz.
- 6- Cevaplarınızı tek bir dosya olarak yükleyiniz.
- 7- yüklediğiniz dosyanın adı, adınızı, soyadınızı ve numaranızı içersin.

Linear Algebra-Final (26.01.2021(13:15-17:00))

Student Name/Number: _____

1. (10) Let P_3 be the set of polynomials of degree ≤ 3 , with real coefficients. Show that the set of polynomials f in P_3 such that $f(2) = 0$ forms a vector space with the standard operations.
2. (10) Prove or disprove that the subset of 2×2 matrices $\mathbf{A}$ such that $\mathbf{A} \begin{bmatrix} 1 & 3 \\ -2 & -6 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ of $\mathbf{M}_{22}$ is a subspace of $\mathbf{M}_{22}$ under the usual matrix operations.
3. (10) Consider the subset $S = \{1 + x^2, x + x^3, 3 - 2x + 3x^2 + 12x^3\}$ of the set of all real coefficient polynomials P , and let $W = \{a + bx + cx^2 + dx^3 | a, b, c, d \in R\}$. Show that $W = \text{span}(S)$.
4. (10) Determine whether the following is linearly independent in $\mathbf{M}_{32}$:

$$\left\{ \begin{bmatrix} 1 & 2 \\ -1 & 1 \\ 3 & 0 \end{bmatrix}, \begin{bmatrix} 4 & 2 \\ -6 & 1 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & -1 \\ 2 & 2 \end{bmatrix}, \begin{bmatrix} 0 & 7 \\ 5 & 2 \\ -1 & 6 \end{bmatrix} \right\}$$

5. (10) Let $U = \text{span}((1, 0, 1, 0, 1), (3, 0, 1, 3, 1), (2, 1, 4, 3, 5), (1, 1, 1, 1, 1), (1, 0, 3, 2, 4))$ be a subspace of R^5 . (i) Find a basis for U ; (ii) Find dimension of U .

6. (15) Find the eigenvalues and corresponding eigenvectors of the matrix $A = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & 1 \\ -1 & 0 & 2 \end{bmatrix}$.

7. (15) $A = \begin{bmatrix} 1 & 0 & -4 \\ 0 & 3 & 2 \\ -2 & 7 & 6 \end{bmatrix}$.

- (a) Find the adjoint matrix of A .
 - (b) Find the determinant of A .
 - (c) Find the inverse matrix of A .
8. (15) Use Cramers Rule to solve the following systems:

$$\begin{aligned} -5x_1 + 2x_2 - 2x_3 + x_4 &= -10 \\ 2x_1 - x_2 + 2x_3 - 2x_4 &= -9 \\ 5x_1 - 2x_2 + 3x_3 - x_4 &= 7 \\ -6x_1 + 2x_2 - 2x_3 + x_4 &= -14 \end{aligned}$$

9. (5) Determine whether each statement is true (T) or false (F).
 - (a) A linearly independent spanning set S is called a basis of a vector space V .
 - (b) Interchanging two rows of a given matrix changes the sign of its determinant.
 - (c) If A is an invertible $n \times n$ matrix, then $Ax = b$ has infinitely many solutions for every b .
 - (d) The set of all first-degree polynomials with the standard operations is a vector space.
 - (e) If an $m \times n$ matrix A is row-equivalent to $m \times n$ matrix B then the row space of A is equal to the row space of B .